Classifying 5M bank documents on a ₹1,00,000 budget

Approach note • IDfy Privy APM take-home • Vansh Gandhi
Code and every number: github.com/vansh-gandhi-02/privy-doc-classifier • Flowchart: separate PDF

Key results

- **Passes the bar on every test set, sealed ones included:** macro-F1 0.982 to 0.998 (bar 0.85); critical recall 0.95 or higher (bar 0.90).
- **₹21,600 for 5M documents** (0.43 paise each), 22% of the budget. Worst plausible case ₹43,800; brute force ₹1,30,170.
- **99.5% of documents never reach the LLM;** about 2,600 (0.05%) go to a person.

The idea

- **Every document carries a probability for each class,** starting from the class mix expected for its file format (never zero).
- **Evidence is bought cheapest first:** metadata, page layout, a few lines of text, the full page, then an LLM. Each step updates the probabilities (Bayes).
- **A document stops as soon as one class reaches 95%;** otherwise it buys the cheapest next step that can still separate the classes in play.
  - A sequential test (Wald's SPRT, many-class form): money goes only where the evidence so far cannot decide.
- **A budget controller moves the gate (τ) mid-run** when spend trends over or under plan.

What we assumed

- **15 classes.** The brief lists 15 and mentions "25" once; we read that as a leftover.
- **Critical: Legal, Property, Proof of Identity (PoI), Proof of Address (PoA).** No class is medical; in a bank, "regulatory" means KYC.
- **Class mix per format:** from typical bank collections; reproduces the brief's format split within 0.03 points. KYC is 20%.
- **Class boundaries:**
  - An ID with a visible address is PoA, otherwise PoI; PAN is always PoI.
  - Anything that transfers, mortgages, titles or values property is Property; every other agreement, rent included, is Legal.
  - Passbooks are not bank statements (labelled Other). An email takes its attachment's class.
- **One file can hold several documents** (merged PDFs); each is classified with its page range.
- **Prices:** the brief's rates; a cloud vCPU taken as 2x slower than our laptop core. **People:** no quoted price, so counted, not costed.
- **Margin:** we target ₹50k of compute (50% margin) with a 10% reserve, so the controller aims at ₹45k.
- **Test data:** synthetic documents for all 15 classes (no real customer data), plus public Indian bank statements (AgamiAI).

How a document moves, cheapest step first

- **L0, open the file** (3 ms): true type, visible text or scan, emails and zips unpacked. Same checksum as an earlier file: copy its label.
- **L1, layout** (0.11 s a page): a frozen document-vision model (DiT) plus six readable signals (page shape, QR code, photo).
  - Settles 64% of scans, all correctly. A passbook and a statement share words, not layout.
- **L2, text:** free text first (1 ms); for scans, the top and largest lines (0.48 s), then the full page (2.2 s), then the next and last pages.
  - A word counts only if it is likelier under one remaining class than another, so shared words cancel out.
  - Checksum-valid IDs (Aadhaar, PAN, GSTIN) are evidence, not verdicts: a loan form quotes a PAN too.
  - Legal and property files open with the same e-stamp page, so they read their later pages first.
- **Address question** (KYC cards, one full read): a valid PIN code plus an address label or state means PoA, otherwise PoI.
- **L3, LLM** (₹0.038 a call): Claude Haiku sees masked text and only the 2 to 5 classes in play plus "none of these".
  - It quotes the deciding words, then answers; the answer is weighed by its measured error rate, not taken as the verdict.
- **L4, a person:** whatever is still below 95% after the LLM.
- **Two safeguards:**
  - **Critical safety net:** a document with a 10% or higher chance of a critical class keeps a 95% gate, whatever the controller does.
  - **Merged PDFs:** split page by page; "Page 3" cannot start a document, "Page 1" always does.

Why the gates sit where they do

- **95% to accept:** an accepted document is wrong at most 5% of the time (measured: 99.9% right). Each step costs several times the last, so stopping early is the cheapest path to the bar.
- **τ, moved by the controller between 85% and 99%,** applies to every gate before the LLM, because reading pages is where the money goes.
  - Measured at 5M: τ 0.85 costs ₹20.7k at 97.9% accuracy; 0.95, ₹21.6k at 99.8%; 0.99, ₹24.3k at 99.6%. Above 95% buys nothing, so it holds 95%.
  - Squeezed to ₹23k with a scan-heavy stretch, it lowered τ to 0.86 and spent ₹23,753, against ₹24,192 without it.
  - The 85% floor is a judgement call: at 80%, too many wrong labels get through for a bank.
- **LLM effort, by a rule fixed in advance:** the cheapest mode whose accuracy clears 95% at the lower 90% confidence bound.
  - On 150 hard documents: answer only, 91.3% (₹0.014 a call); quote first, 98.7% (₹0.038). Quoting wins, for about ₹700 more across 5M.

Local intelligence

- **Everything except the LLM runs on CPU:** a pretrained layout model (DiT) with a classifier trained in seconds; a text model over 3 to 5 letter pieces (survives OCR typos); checksummed Indian IDs; the address rule, page splitter and masker.

- **Every model is calibrated,** so a reported 95% is right 95% of the time. Together they settle 99.5% of documents before any LLM call.

Results on synthetic documents (documents sent to a person count as errors)

Test set	Macro-F1	Critical recall	LLM / person
Dev-test: 843 documents	0.998	1.00 on all four	0.4% / 0.1%
Lockbox A: 881 documents, 40% in unseen layouts; sealed, run once	0.990	PoI 1.00, PoA 0.98, Legal 0.95, Property 1.00	10 / 6
Lockbox C: 430 fresh documents, built after everything was frozen; run once	0.982	1.00 on all four	3 / 1
Lockbox D: 120 outside bank statements; run once	120/120 kept whole; 114/120 labelled bank statement		26 / 4

- **No lockbox was used to train, tune or choose anything.** Each was run once.
- **One weakness found and fixed:** on Lockbox B (120 other outside statements) the splitter cut each statement into 2.5 pieces. Reading printed page numbers fixed it (16/80 kept whole before, 80/80 after, on 80 further statements), confirmed once on Lockbox D.

Cost at 5M documents

Item	₹
Open and look at every page (L0 + L1)	2,343
Selective reads of scanned pages, and their page-number lines	9,809
Further reads (full pages, later pages, KYC address reads)	8,280
LLM (0.6% of documents)	1,169
People (about 2,600 documents; no price quoted)	0
Total (0.43 paise a document; 22% of the budget)	**21,601**

- **Measured precision: ±5.4% at 95% confidence** (₹20,449 to ₹22,766), from timings, per-class rates and token counts.
- **Assumptions move it more:** cloud vCPU 1x to 3x slower, ₹11.4k to ₹31.8k; 1.7 to 6.2 pages per PDF, ₹18.4k to ₹29.6k; worst case ₹43,794, under the ₹50k target.
  - Both are cheap to pin down: an L0 pass over the bucket (about ₹225) gives page counts; timing the first 1,000 documents gives cloud speed.

Limitations

- **Accuracy is on synthetic documents.** IDs and marksheets follow official designs but were not checked against real copies; bank statements are the only outside source tested.
- **The masker misses names.** On a public Indian PII set it removes PAN, licence, email and UPI 100%; Aadhaar 88%; phone 85%; date of birth 62%.
- **A real run needs a labelled seed:** the LLM labels about 5,000 documents; people check only where it disagrees with our models (about 600).

Next steps

- **A name masker.**
  - **Why:** names are the one personal detail the LLM still sees.
  - **Why not yet:** it needs a trained model for Indian names; patterns cannot catch them.
- **A small fine-tuned text model (LoRA) for L2.**
  - **Why:** it reads meaning, so fewer documents reach the LLM and people.
  - **Why not yet:** it needs labelled real documents; on synthetic data the current model already settles 99.5%, so a gain is unmeasurable.
- **Conformal prediction on critical classes.**
  - **Why:** turns “recall of 0.90 or more” from a measured result into a guarantee.
  - **Why not yet:** it is only as good as its calibration set; on synthetic data it would describe our generator, not the bank.
- **A vision LLM for scans with no readable text.**
  - **Why:** these are the documents most likely to reach a person today.
  - **Why not yet:** a page image cannot be masked like text, so full ID numbers and photos would reach the LLM.
- **More outside document types for the splitter.**
  - **Why:** bank statements were the only outside type tested.
  - **Why not yet:** no safe public Indian KYC, deed or marksheet sets exist; search hits were real people’s documents.